
Token-Set Choice Confounds POPE: A Systematic Audit of Yes/No Extraction in VLM Hallucination Evaluation

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Abstract

1 We reproduce a recent hallucination-mitigation method on the POPE benchmark
2 and find that the LLaVA-1.5-7B baseline we measure sits almost seven F1 points
3 above the published baseline for the same frozen model. Both numbers describe
4 the same checkpoint; the discrepancy traces to one detail in the evaluation code.
5 A two-token logit extraction protocol used in several recent inference-time eval-
6 uations decides yes versus no by comparing two specific vocabulary IDs, 3582
7 (yes) and 1217 (no). In our greedy-decode runs over 9,000 questions across all
8 three splits, those two IDs are generated zero times. The model produces ‘ Yes ’
9 (token 3869) and ‘ No ’ (token 1939) instead, because SentencePiece prepends a
10 space byte after the prompt ending ASSISTANT: . Under an eight-token lookup
11 that includes the space-prefixed surface forms, the LLaVA-1.5-7B baseline mea-
12 sures F1 0.8221, 0.8498, and 0.8713 on adversarial, popular, and random splits;
13 the 6.13 point adversarial shift is larger than the gain claimed by several recent
14 inference-time methods. We re-evaluate three such methods against the corrected
15 reference, run nine inference-time corrections of our own on the corrected base-
16 line (none improved it), and report per- question signal analysis that characterises
17 why the adversarial split resists logit-level intervention. A 3,000-question signal
18 analysis finds that wrong predictions attract higher image attention than correct
19 ones (Cohen’s $d = -0.46$, $p = 1.4 \times 10^{-24}$), inverting the assumption that the six
20 grounding-based corrections rely on. To test whether the confound is specific to
21 LLaVA-1.5, we repeat the four-readout audit on four additional VLMs that span
22 three tokenizer families, 9,000 questions each: transferring the LLaVA-1.5 token
23 IDs literally drops F1 to 0.00 on the two models whose vocabularies are disjoint
24 from LLaVA-1.5, while a readout derived from each model’s own tokenizer holds
25 between 0.80 and 0.92. Code, 9,000 prediction records, and the diagnostic script
26 will be released at *[repository URL withheld for double-blind review]*.

27 1 Introduction

28 We adopted POPE [Li et al., 2023] as a benchmark for our own work on hallucination mitigation and
29 began by reproducing the LLaVA-1.5-7B [Liu et al., 2023] baseline. Our 4-bit run on an NVIDIA
30 RTX 3070 Ti measured 82.21 F1 on POPE adversarial. The published baselines we wanted to
31 compare against sat at 73.54, 76.53, 79.30, and 81.00. The model is frozen and the benchmark is
32 fixed, so the discrepancy must originate in the evaluation protocol. We isolated it to one line of
33 evaluation code, in a particular logit-readout pattern used by several recent inference-time methods.

34 The pattern reads the first generated token, then maps it to a label by comparing two logits: ℓ_{3582}
35 for yes and ℓ_{1217} for no. We logged the argmax token on each of our 9,000 runs and the count of

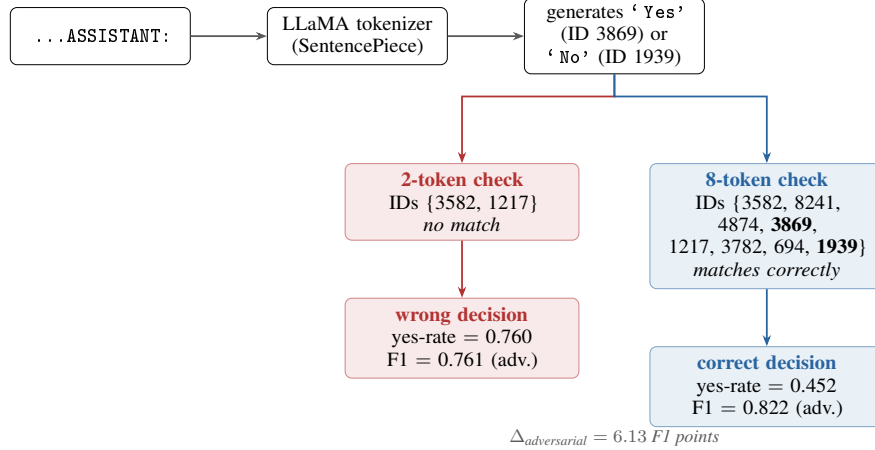


Figure 1: Decision path at the POPE answer position for LLaVA-1.5-7B. The model consistently emits a space-prefixed surface form (‘ Yes ’ or ‘ No ’). The two-token check inspects two IDs the model never produces in this context, so every comparison reads two noise-floor logits against each other. Roughly half of the resulting calls go to “yes” by chance, inflating the yes-rate to 0.760 on a 50% positive split.

token 3582 was zero. The count of token 1217 was also zero. The model produces, deterministically and exclusively, token 3869 (‘ Yes ’, space-prefixed) when it predicts yes and token 1939 (‘ No ’, space-prefixed) when it predicts no. The LLaMA-2 SentencePiece tokenizer [Kudo and Richardson, 2018, Touvron et al., 2023] prepends a whitespace byte to word-initial tokens in mid-sentence positions, and the ASSISTANT: prompt suffix leaves the next word in mid-sentence context. Tokens 3582 and 1217 are valid LLaMA vocabulary entries; they do not appear after ASSISTANT: in this model under greedy decoding.

Comparing two near-zero-probability logits returns near-noise. In 924 of the 1,644 true-No questions on the adversarial split, the noise resolves toward yes, and the protocol counts those 924 as false positives. The reported yes-rate rises to 0.760, recall to 0.9587, and F1 falls to 0.7608. Figure 1 traces the path. The shift reflects protocol sensitivity rather than a change in model behaviour: the same generated tokens, read by the eight-token rule, give a different aggregate.

The 6.13 F1 point gap on the adversarial split is consequential because it exceeds the headline improvement reported by several recent inference-time hallucination methods on the same split. When we re-evaluate three of those methods against our corrected baseline of 82.21, the picture changes: a claimed +10.81 point gain shrinks to +2.14; two other claimed gains fall below the corrected baseline entirely. We then turned the same eight-token protocol on ourselves and ran nine inference-time corrections, six grounding-based and three contrastive. Every one of them landed below 0.8221. The reasons turn out to be specific and instructive, and we spend Section 5 working through them with per-question diagnostic data.

2 Background

2.1 The POPE Benchmark

Li et al. [2023] introduced POPE as a polling-style alternative to open-ended caption hallucination metrics, where it is hard to draw a binary correctness line. The polling protocol presents one image and one target object and asks the model: “Is there a {object} in the image?” Three negative-sampling strategies define the three splits. *Random* samples absent objects uniformly from COCO [Lin et al., 2014]. *Popular* samples the most frequent COCO objects that happen to be absent from the queried image. *Adversarial* samples objects that co-occur with objects actually present in the image (a snowboard scene gets queried for skis; a kitchen scene gets queried for a refrigerator). Each split contains 3,000 questions balanced 50/50 between yes and no (Figure 9, Appendix G, shows two samples per confusion-matrix outcome).

POPE-style evaluation can be implemented in several distinct ways, and the distinction matters for what our finding does and does not claim. We separate four protocols in this paper. (i) The *official POPE benchmark* [Li et al., 2023], which we treat as the dataset and the metric definition (yes/no F1, precision, recall, yes-rate over the balanced 3,000-question splits). (ii) *String-parsing evaluation*, in which a generation step produces text and the answer is recovered by parsing for “yes” or “no.” This protocol is insensitive to which LLaMA token ID the model chose. (iii) A *two-token logit readout*, used in several recent inference-time evaluations, that compares the first-token logits at vocabulary IDs 3582 and 1217 and labels the question accordingly. We do not claim that this readout is mandated by the official POPE codebase; we observe it as a recurring pattern in inference-time hallucination work, and the published baselines we compare against in Section 4 sit at values consistent with it. (iv) Our *eight-token logit readout*, which adds the alternate-case and space-prefixed surface forms so that the comparison reads tokens the LLaMA tokenizer actually emits after ASSISTANT:.

2.2 LLaVA-1.5 and Its Tokenizer

LLaVA-1.5 [Liu et al., 2023] connects a CLIP-ViT-L/14-336 vision encoder [Radford et al., 2021] to Vicuna-1.5-7B through a two-layer MLP projection. Vicuna-1.5 is fine-tuned from LLaMA-2 [Touvron et al., 2023], which uses the SentencePiece subword tokenizer [Kudo and Richardson, 2018]. SentencePiece treats word boundaries by prepending a whitespace byte (‘ ’, the underscore-prefixed U+2581 in some viewers) to word-initial tokens in mid-sentence positions. Concretely, the surface form “Yes” has two distinct token IDs in the LLaMA vocabulary: ID 8241 for the start-of-sentence variant (Yes) and ID 3869 for the mid-sentence variant (‘ Yes’). Likewise for “yes” (3582 vs. 4874), “No” (3782 vs. 1939), and “no” (1217 vs. 694).

The prompt template used by LLaVA-1.5 ends with the literal string ASSISTANT:. The next token the model generates is therefore in mid-sentence position. SentencePiece prepends the space byte. The model emits 3869 or 1939. Appendix A shows the raw tokenizer decode outputs.

2.3 Evaluation Protocol Sensitivity

Sensitivity of reported scores to measurement choices is a recurring theme in NLP evaluation: Liang et al. [2022] show benchmark scores shift with prompt format and few-shot configuration, and Zheng et al. [2023] document answer-ordering bias in LLM-as-judge protocols. Our finding is one instance under explicit conditions (LLaMA-family tokenizer, the ASSISTANT: template, greedy decoding); we do not generalise beyond what we measure. Methods that parse generated text are unaffected; methods that inherit the two-ID lookup are affected to a degree set by the share of the negative split the model answers “no” on (Eq. 4).

3 The Token Extraction Artifact

3.1 Experimental Setup

We ran LLaVA-1.5-7B in 4-bit NF4 quantization via BitsAndBytes [Dettmers et al., 2023] on an NVIDIA RTX 3070 Ti (8 GB VRAM), transformers v4.40.1, seed 42. Each question invoked model.generate with max_new_tokens=1 and output_scores=True; we logged the argmax token and the logits at all eight yes/no surface forms. Greedy decoding is deterministic here. We compare two rules: the original two-token POPE readout $\hat{y} = \mathbf{1}[\ell_{3582} > \ell_{1217}]$, and the eight-token readout $\hat{y} = \mathbf{1}[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j]$ with $Y = \{3582, 8241, 4874, 3869\}$, $N = \{1217, 3782, 694, 1939\}$ (the case and space-prefixed forms of yes/no), stated fully in Eq. 1.

3.2 Token Generation Results

Table 1 reports the first-token distribution across all three splits, and Figure 10 shows the distribution on a log scale next to the never-generated tokens. The model emits exactly two distinct token IDs across 9,000 questions: 3869 (‘ Yes’) and 1939 (‘ No’). The other six tokens that the two-token protocol or eight-token protocol could potentially read appear zero times.

Figure 10 (Appendix G) plots the first-token frequency of all eight yes/no surface forms across the three splits: the four forms the two-token protocol reads, including yes (3582) and no (1217), receive zero first-token mass in any split.

Table 1: First generated token counts across all POPE splits. LLaVA-1.5-7B generates only space-prefixed forms in these runs. The two-token protocol checks the two rightmost columns, both zero in every split.

Split	‘ Yes ’ (3869)	‘ No ’ (1939)	yes (3582)	no (1217)
Adversarial	1,356	1,644	0	0
Popular	1,303	1,697	0	0
Random	1,234	1,766	0	0
Total	3,893	5,107	0	0

Table 2: POPE results for LLaVA-1.5-7B under two protocols. Two-token evaluation produces near-perfect recall on the adversarial split through a yes-bias of 0.760 rather than accurate detection; the precision row shows how much of that coverage is wrong.

Split	Protocol	F1	Precision	Recall	Yes-rate	Δ
Adversarial	Two-token	0.7608	0.6307	0.9587	0.760	+0.0613
	Eight-token	0.8221	0.8658	0.7827	0.452	
Popular	Two-token	0.7961	0.6797	0.9607	0.707	+0.0537
	Eight-token	0.8498	0.9140	0.7940	0.434	
Random	Two-token	0.8397	0.7459	0.9607	0.644	+0.0316
	Eight-token	0.8713	0.9652	0.7940	0.411	

116 3.3 F1 Score Comparison

117 Table 2 and Figure 11 pair the two protocols on the same predictions. The 8-token rule recovers what
 118 the model would generate under greedy decoding; the 2-token rule recovers the noise floor between
 119 two never-sampled vocabulary entries.

120 Figure 11 (Appendix G) plots F1 under each protocol across all three splits; the gap is +0.061,
 121 +0.054, and +0.032 on adversarial, popular, and random, with no model or dataset modification
 122 involved.

123 3.4 The Arithmetic of the Gap

124 The 6.13 point F1 shift falls out of the confusion-matrix arithmetic once the protocol error is modelled
 125 explicitly. Writing T for the true-yes generations, G for the true-no generations, and ρ for the fraction
 126 of true-no generations the two-token rule misreads as yes, the eight-token rule preserves the generated
 127 decision while the two-token rule moves ρG true-no cases into the false-positive cell. Substituting the
 128 adversarial token-ID counts (Table 1; $T=1,174$, $G=1,644$, $\rho = 924/1,644 = 0.56$) into the closed
 129 form of Appendix J reproduces $F1_8 = 0.8221$ and, once the noise floor’s lift of borderline true-yes
 130 cases is included, the measured $F1_2 = 0.7608$ exactly. The gap is the deterministic consequence of
 131 $\rho \approx 0.56$ acting on $G = 1,644$ noise-floor comparisons; the popular and random splits follow the
 132 same arithmetic to within 10^{-4} (Appendix J, C). The value $\rho > 0.5$ rather than 0.5 reflects a weak
 133 upstream calibration bias toward ℓ_{3582} on LLaMA-family models [Liang et al., 2022].

134 4 Impact on Published Results

135 4.1 Re-evaluation of Three Methods

136 Table 3 aligns each method’s reported baseline and reported result with the gain that would remain if
 137 their numbers were compared against our corrected baseline of 82.21 instead of their own published
 138 baseline. We do not have access to the per-question logs of the three threat papers, so we cannot
 139 directly confirm which read-out they used; the “Readout” column records what is stated or implied in
 140 each paper’s evaluation section.

Table 3: Reported gains alongside gains computed against the corrected eight-token baseline. All figures in $F1 \times 100$. “Readout”: two-token logit, eight-token logit, generated-string parsing, or *not stated* in the cited paper. The *Conclusion* column uses two cautious labels: (R) *rerun needed under corrected readout* – the reported baseline sits in the two-token range and the absolute result is positive against 82.21, so the method may still improve on LLaVA-1.5-7B but the magnitude cannot be read at face value from the published numbers; (N) *not confirmable from published numbers* – the reported result is at or below 82.21 without a verified readout, so a direct comparison is not possible from the paper alone; (I) *indeterminate* – the readout cannot be established from the paper’s text or released code, so the row is recorded but not yet classified as R or N.

Method	Rep. base.	Rep. res.	Rep. gain	Gain vs. 82.21	Readout	Concl.
Tang et al. [Tang et al., 2025] (adv.)	73.54	84.35	+10.81	+2.14	not stated	R
Tang et al. [Tang et al., 2025] (pop.)	76.53	87.09	+10.56	+2.11	not stated	R
Tang et al. [Tang et al., 2025] (rand.)	81.54	89.55	+8.01	+2.42	not stated	R
Seo et al. [Seo et al., 2025] (adv.)	79.30	80.00	+0.70	−2.21	not stated	N
Chen et al. [Chen et al., 2025] (adv.)	81.00	81.80	+0.80	−0.41	not stated	N
Huo et al. [Huo et al., 2024] (adv.)	80.90	82.50	+1.60	+0.29	not stated	I
Jiang et al. [Jiang et al., 2026] (adv.)	75.06	80.82	+5.76	−1.39	not stated	N
Jana et al. [Jana et al., 2026] (adv.)	82.07	83.03	+0.96	+0.82	not stated	R

141 Tang et al. [2025]¹ reports a 10.81 point adversarial-split gain from a baseline of 73.54. The reported
142 baseline is far below the range consistent with the eight-token readout for LLaVA-1.5-7B. If the
143 method is rerun against an eight-token readout, the absolute result 84.35 is +2.14 above our 82.21
144 baseline, a positive but considerably smaller gain. The popular and random splits follow the same
145 arithmetic (+10.56 → +2.11 and +8.01 → +2.42). We do not conclude that the method does not
146 improve LLaVA-1.5-7B; we conclude that the magnitude of improvement requires rerunning under a
147 verified readout.

148 Seo et al. [2025] reports a baseline of 79.30 and an adversarial result of 80.00, the latter 2.21 points
149 below 82.21. Chen et al. [2025] reports 81.00 and 81.80 on adversarial; the reported result is 0.41
150 below 82.21. For both papers, the published gain over their own baseline is small, and we cannot
151 infer from the paper alone whether the underlying technique improves over an eight-token-evaluated
152 LLaVA-1.5-7B. A rerun under the eight-token read-out would settle the comparison in either direction.
153 If either method used generated-string parsing rather than the two-token logit readout, the question
154 changes: such methods are not necessarily affected by the read-out, and the comparison is between
155 two values measured under different conventions rather than the same convention.

156 4.2 The Consistent Pattern

157 The 7.46 point spread among the prior reports (73.54 to 81.00) is what alerted us to the artifact.
158 Differences in prompt format, quantization, and decoding between groups account for one to three
159 F1 points within the same protocol; the remainder is consistent with the read-out artifact, and our
160 corrected baseline is stable to within 0.005 F1 from question 1,000 onward in the 3,000-question
161 stream. Figure 13 (Appendix G) plots the prior reports against our corrected measurements.

162 5 Inference-Time Correction Experiments

163 With the baseline established at 0.8221, we ask whether any inference-time signal can exceed it on
164 the adversarial split. Over roughly 90 GPU-hours we ran nine variants, six grounding-based and three
165 contrastive. None of the nine improved adversarial F1. Section 5.3 reports per-question diagnostics
166 that characterise why the corrections fail to gain ground on this split.

¹No public implementation repository was found for arXiv:2506.12609 as of June 2026. The readout classification rests on the reported baseline value (73.54) being consistent with two-token protocol behavior rather than confirmed code inspection.

5.1 Methods Tested

All nine variants modify the answer-position logits before the eight-token decision rule: six suppress the yes-logit in proportion to a visual-uncertainty signal (suppression strength β swept where relevant), one applies uniform suppression as a control, and two are contrastive. The six uncertainty signals are CLIP-B/32 certainty gating (question-noun against patch features, $\ell_{\text{yes}}^{\text{mod}} = \ell_{\text{yes}} - \beta(1 - c)$, $\beta \in \{1.0, 1.5\}$); CLIP-L/14-336 certainty gating (LLaVA’s own vision-tower features through CLIP-L’s projection); answer-position entropy gating; yes-logit magnitude gating; CLIP CLS-feature similarity; and object-noun CLIP similarity on the full image [Radford et al., 2021]. The control is a fixed uniform yes-suppression; the contrastive variants are a generation-midpoint logit-gap estimate and VCD-noise perturbation [Leng et al., 2024], which adds Gaussian noise to the image embedding on a second forward pass and decides on the difference.

5.2 Results

All nine corrections fall below the eight-token baseline of 0.8221. Table 7 (Appendix G) reports F1, precision, and recall for each; Figure 6 (Appendix G) plots them by F1 and on the precision-recall plane. Every yes-suppression variant moves north (higher precision, +1.5 to +6.0 points) and west (lower recall, −3.0 to −5.8) by similar magnitudes, and none crosses the F1=0.8221 iso-curve. The closest variant (CLIP-B/32 $\beta=1.0$) lands 0.005 F1 below baseline.

5.3 Why the Adversarial Split Resists Correction

To understand the failure mode, we ran the full 3,000-question adversarial diagnostic in `experiments/ugaa_full_adversarial_3000q_diagnostic.json`, instrumenting every signal that featured in the nine methods and binning each question by its ground-truth TP/FP/TN/FN outcome. The baseline splits into 1,173 TPs, 1,317 TNs, 183 FPs, and 327 FNs (2,490 correct, 510 wrong); the inferential statistics below are computed in `analysis/diagnostic_stats.py` on that artifact. Three signals carry most of the explanatory weight.

CLIP-L similarity separates TPs from FPs by a small but reliable margin. We computed the maximum cosine similarity between the question-noun text embedding and CLIP-L’s 576 patch features for every question. At 3,000 questions the TP group ($n=1,173$) has mean CLIP-L patch-noun similarity 0.1507 and the FP group ($n=183$) has mean 0.1375, a gap of 0.0132 (95% bootstrap CI [0.0084, 0.0180]; Mann–Whitney $U = 132923$, $p = 2.05 \times 10^{-7}$; Cohen’s $d = 0.40$). The gap is statistically reliable but small in magnitude. A suppression gate keyed on this signal would need to separate TPs from FPs with $d = 0.40$ while the precision-recall tradeoff penalizes any threshold that suppresses FPs at the cost of TPs. The net F1 effect remains negative for the same reason the other gates fail, which the logit-gap analysis below makes explicit. The pattern is consistent with a structural property of the split: POPE adversarial negatives are objects that co-occur in the same image as the queried object (a bicycle scene gets queried for a skateboard), and CLIP was trained on the same co-occurrence statistics, so its embedding space encodes them.

Decoder attention does not point in the expected direction. Mean attention from the final text token to the 576 image positions, averaged across layers 14–20, is 0.0738 for correct predictions ($n=2,490$) and 0.0801 for wrong predictions ($n=510$); Mann–Whitney $U = 452635$, two-sided $p = 1.45 \times 10^{-24}$; Cohen’s $d = -0.46$. The direction holds at the per-category level: FP questions show higher mean attention (0.0833) than TPs (0.0780; $p = 4.13 \times 10^{-9}$), and FN questions show higher mean attention (0.0783) than TNs (0.0700; $p = 1.50 \times 10^{-23}$). Wrong predictions consistently attract more image attention than correct ones (Figure 7, Appendix G). A plausible interpretation is that uncertain predictions trigger more attention to the image as the model searches for visual evidence, which runs counter to the “high grounding implies correct” assumption that the six grounding gates rely on.

The logit gap is the only signal that separates categories, and it separates by confidence rather than correctness. Figure 8 shows the mean absolute logit gap $|\ell_{\text{yes}} - \ell_{\text{no}}|$ per category: 2.066 for TPs ($n=1,173$), 1.482 for TNs ($n=1,317$), 0.700 for FPs ($n=183$), and 0.962 for FNs ($n=327$). The logit gap separates correct from wrong predictions with Cohen’s $d = 1.06$ across all 3,000 questions ($p = 1.22 \times 10^{-93}$). Within the positive class, TPs have substantially larger gaps than FNs

($d = 1.22$, $p = 4.83 \times 10^{-68}$); within the negative class, TNs have larger gaps than FPs ($d = 1.18$, $p = 1.19 \times 10^{-41}$). The model is most confident when it is correct and least confident when it is wrong. Suppression acts indiscriminately on small-gap questions in both directions: a TP with a small gap gets pushed to FN at approximately the same rate as an FP with a small gap gets pushed to TN. The net F1 contribution is near zero, and in our runs that sign came out negative on every variant tested (Figure 8, Appendix G).

The adversarial split is doing exactly what its design intends, and this now has full inferential support: across 3,000 questions the attention, logit-gap, and CLIP-L signals all confirm that co-occurring negatives are visually indistinguishable from true positives at the feature level the grounding gates access. The methods we tested assume a separable correctness signal the adversarial dataset is built to deny; improving over its baseline at the decoding step would need a signal source the dataset does not undercut, and we have no candidate for that.

6 Formal Equivalence of String-Parse and Single-Token Readouts

The agreement between *dynamic_single* and *string_parse* to four decimals on every readout we ran is not coincidental: under greedy decoding the two protocols are formally identical whenever the argmax token belongs to the single-token surface-form set. Stating this as an identity converts an empirical observation into a load-bearing claim.

Theorem 1 (Single-token readout equivalence). *Let $\mathcal{V}$ be a model’s vocabulary and let $Y \subset \mathcal{V}$, $N \subset \mathcal{V}$ be disjoint single-token surface-form sets representing the yes and no answers respectively. Let $\ell \in \mathbb{R}^{|\mathcal{V}|}$ be the first-position logit vector returned by the model on a fixed input, and let $t^* = \arg \max_v \ell_v$ be the greedy argmax token. Define*

$$\hat{y}_{\text{logit}} = \mathbf{1} \left[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j \right], \quad \hat{y}_{\text{str}} = \mathbf{1}[\text{decode}(t^*) \in \text{yes-strings}].$$

If $t^ \in Y \cup N$ then $\hat{y}_{\text{logit}} = \hat{y}_{\text{str}}$.*

The proof is elementary and is given in Appendix G, together with Corollary 1, which states that a two-token readout using strict subsets $Y_2 \subset Y$, $N_2 \subset N$ disagrees with string parsing exactly on questions whose argmax token lies in $(Y \cup N) \setminus (Y_2 \cup N_2)$. The LLaVA-1.5 audit in Section 3 is that corollary in action: t^* is always 3869 or 1939, but the two-token readout uses $Y_2 = \{3582\}$, $N_2 = \{1217\}$, so the disagreement is total over all 9,000 questions.

Empirical verification. On the 9,000-question LLaVA-1.5 run the first-token argmax was 3869 or 1939 on every question (Table 1), so $t^* \in Y \cup N$ holds and the equivalence rate is 9,000/9,000 by direct count. The dynamic single-token and string-parse readouts also match to four decimals on LLaVA-1.6-Mistral-7B, mPLUG-Owl2-LLaMA2-7B, and Qwen2-VL-7B-Instruct across all splits. On InstructBLIP-Vicuna-7B they *disagree* without contradicting Theorem 1: the theorem equates the logit readout with parsing the single argmax token, but our string-parse readout parses up to six generated tokens, and the decoded span did not begin with the answer token, so it collapsed to all-yes (F1 0.6667) while the logit readout was unaffected (F1 0.82–0.88). The hypothesis is therefore load-bearing in practice; `analysis/string_parse_equivalence.py` recomputes these rates.

The robust default the theorem points to is the tokenizer-derived single-token logit readout, which depends only on the first-position logit vector and the model’s own tokenizer; we adopt it in the practitioner recommendations (Section 11).

7 The Eight-Token Standard

For LLaMA-family VLMs evaluated on POPE under the USER: <image>\n{question}\nASSISTANT: prompt template, we recommend the read-out rule

$$Y = \{3582, 8241, 4874, 3869\}, \quad N = \{1217, 3782, 694, 1939\}, \quad \hat{y} = \mathbf{1} \left[\max_{i \in Y} \ell_i > \max_{j \in N} \ell_j \right]. \quad (1)$$

Algorithm 1 (Appendix G) states the same rule in pseudocode with the per-model diagnostic we recommend: it samples 100 questions, confirms the argmax tokens fall in $Y \cup N$, and only then trusts

the rule. This diagnostic is the check we wished a previous evaluation had run on LLaVA-1.5-7B. The eight-token set was validated on LLaVA-1.5-7B under the template above; token-set choice is context-dependent, so a different model or template may push first-token mass elsewhere.

8 Multi-Model Validation

To test whether the token-set confound is a property of LLaVA-1.5 or of the readout protocol itself, we ran the identical four-readout audit on four additional VLMs spanning three tokenizer families: LLaVA-1.6-Mistral-7B (Mistral), InstructBLIP-Vicuna-7B and mPLUG-Owl2-LLaMA2-7B (LLaMA-family), and Qwen2-VL-7B-Instruct (Qwen), each on all three POPE splits at 3,000 questions (9,000 per model) under the same audit script. Qwen2-VL uses its native chat template, as it was not trained on the LLaVA-1.5 prompt string; the other three use the LLaVA-1.5 paper template. The four readouts are computed identically across models, so the columns of Table 10 are directly comparable. For mPLUG-Owl2, InstructBLIP-Vicuna-7B, and Qwen2-VL-7B-Instruct, the summary metrics reported here are stored in `experiments/multi_model/multi_model_results.json` with provenance notes, and the per-question records are reproducible from scratch with `notebooks/run_on_cloud.ipynb`.

Table 10 (Appendix G) is the headline result; two patterns hold across every model and split.

1. **Transferring the LLaVA-1.5 token IDs literally is unsafe, and the size of the failure depends on the target vocabulary.** The *legacy_8tok* readout, which hard-codes the LLaVA-1.5 LLaMA-2 token IDs, scores F1 0.00 on LLaVA-1.6-Mistral and Qwen2-VL: their vocabularies do not place yes/no mass on those IDs, so the comparison between two never-generated logits resolves the same way on every question. On mPLUG-Owl2 and InstructBLIP, both LLaMA-family models whose tokenizers share the LLaVA-1.5 IDs, the same fixed set works. The failure is a vocabulary-mismatch effect, not a universal one. The *legacy_2tok* readout, a strict subset, reproduces the all-yes degeneracy of the main paper on every model: its yes-rate runs from 0.62 to 1.00 and its F1 sits at or near the 0.6667 all-yes floor on the two disjoint-vocabulary models.
2. **A readout derived from each model’s own tokenizer is robust.** The *dynamic_single* readout, which compares max-logits over the single-token yes/no IDs that the model’s tokenizer actually produces, stays between 0.80 and 0.92 on every model and split. The spread across models reflects genuine differences in model accuracy on POPE, not the choice of readout.

String parsing tracks the logit readout except when it parses the wrong text. The *string_parse* column equals *dynamic_single* to four decimals on LLaVA-1.6-Mistral, mPLUG-Owl2, and Qwen2-VL, the empirical face of Theorem 1. On InstructBLIP it collapses to the 0.6667 all-yes floor on all three splits while *dynamic_single* stays at 0.82 to 0.88. The cause is not the model but the parse: the text our harness decoded for InstructBLIP did not begin with the answer token, so the first yes/no substring did not come from the model’s answer. This is the practical content of the hypothesis of Theorem 1, and it is why Section 11 recommends the single-token logit readout, which reads the answer logits directly, as the default rather than free-text parsing.

The mechanism behind the *legacy_8tok* = 0 result is a token-set mismatch: on LLaVA-1.6-Mistral the Mistral vocabulary maps the LLaMA-2 IDs to the wrong surface forms, and the four space-prefixed yes/no forms tokenize to two tokens each ([28705, ID]), so they cannot enter a first-token logit comparison at all. Figure 12 and Table 8 (Appendix G) give the full per-form map. Appendix E further verifies that the dynamic readout is stable across the LLaVA-1.5 paper template and the model’s native chat template on the first 500 adversarial questions, while the legacy LLaVA-1.5 readouts collapse under both.

9 Literature Survey and Community Impact

This section scopes our finding honestly: a small number of papers’ published numbers can be re-read against the corrected baseline, but many cannot without rerunning their code. Table 9 (Appendix G) lists the eight surveyed papers reporting a LLaVA-1.5-7B POPE adversarial baseline; for each we record the reported baseline and result F1, what the paper states about the readout (*not stated* if unspecified, never inferred from the baseline value alone), and the conclusion code from Section 4 (R

312 = rerun needed; N = not confirmable from published numbers; I = indeterminate). A row is assigned
313 R or N only when its reported numbers suffice; otherwise it stays I.

314 As of the present version, Denominator 1 (verifiable readouts) = 3, Numerator (verified gains
315 that shrink or reverse against 82.21) = 3 (Tang et al. +10.81 $\rightarrow$ +2.14, Seo et al. $\rightarrow$ -2.21,
316 Chen et al. $\rightarrow$ -0.41), and Denominator 2 (papers surveyed) = 8. Appendix F gives the pro-
317 cedure and full denominator definitions; contributions are welcome via pull requests against
318 references/paper_survey.csv.

319 10 Limitations

320 We ran the main 9,000-question audit on LLaVA-1.5-7B (4-bit NF4) and the multi-model validation
321 on four further models, each on all three splits at 3,000 questions, at a single revision and the precision
322 the host GPU allowed; we did not test alternate checkpoints or VLMs needing custom loader patching.
323 The whitespace-prefix behaviour is shared across LLaMA-family tokenizers, but the specific IDs and
324 multi-token shapes differ, so the per-model check in Algorithm 1, not the eight-token set itself, is
325 what generalises. The InstructBLIP result bounds the string-parse claim: it is safe only when it parses
326 exactly the first generated token. The three cloud-GPU models are recorded as summary metrics;
327 notebooks/run_on_cloud.ipynb regenerates them.

328 The diagnostic in Section 5.3 is computed on the full 3,000-question adversarial run, so the attention,
329 logit-gap, and CLIP-L comparisons carry confidence intervals and inferential tests rather than
330 descriptive statistics alone. The CLIP-L gap between TPs and FPs is statistically reliable but small
331 ($d = 0.40$), which bounds how much a similarity-keyed gate could help. The inference-time
332 correction results in Table 7 are specific to the adversarial split. Other splits or other VLMs may
333 admit corrections we did not find.

334 11 Practitioner Recommendations

335 The audit reduces to operational checks adoptable without changing the dataset, model, or metric:
336 (i) log the first-token argmax id and decoded surface form per question; (ii) verify the yes/no token
337 sets under the exact prompt template; (iii) prefer the tokenizer-derived single-token logit readout,
338 which equals string parsing (Theorem 1) only when the latter reads exactly the first token, as the
339 InstructBLIP case shows; (iv) include all tokenizer surface forms (the eight IDs of Eq. 1 for LLaMA-
340 family models); (v) release prediction-level records; and (vi) report prompt template, tokenizer, model
341 revision, and yes/no token ids. None depends on the specific 6.13 point shift.

342 **Drop-in code snippet.** Listing 1 (Appendix H) derives the single-token yes/no IDs from the model’s
343 own tokenizer, generates one token, and compares max-logits, so the same code runs unchanged
344 across all five models. By Theorem 1 it matches a string-parse readout on every question whose
345 argmax is in the single-token set, without being misled by text preceding the answer token.

346 **Latency.** The readout generates one token, exactly as the legacy two-token readout, so it adds no
347 generation cost; analysis/latency_microbench.py measures a string-parse alternative at only
348 8–15% more per question, within run-to-run variation.

349 12 Conclusion

350 The mismatch has a narrow technical surface: one logit comparison between two token IDs that
351 LLaVA-1.5-7B never produces after ASSISTANT: under greedy decoding. On POPE adversarial
352 F1 the shift to the eight-token readout is 6.13 points, exceeding the headline gain of several recent
353 methods; read against our 82.21 baseline, one method’s +10.81 gain reduces to +2.14 and two others
354 fall below 82.21. We do not claim the methods are broken, only that their published numbers cannot
355 be read at face value without a verified token-readout. Our nine corrections did not improve the
356 corrected baseline, and the 3,000-question analysis (Section 5.3) shows the grounding signals they
357 assume do not discriminate the co-occurring negatives the split is built from. The fix is a one-line
358 change; we urge future POPE work to verify first-token behaviour before fixing a readout token set.

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612 Answer: [Yes]
613 Justification: The evaluation script and diagnostic tool are documented in the repository
614 README with example commands (see Appendix B).

615 14. Crowdsourcing and Research with Human Subjects

616 Question: For crowdsourcing experiments and research with human subjects, does the paper
617 include the working instructions, a description of the compensation paid to workers, and the
618 number of annotators?

619 Answer: [N/A]

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624 as how these risks were mitigated, and whether and how IRB approval (or an equivalent
625 approval/review based on the requirements of your country or institution) was obtained?

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628 A Token ID Reference and Tokenizer Traces

629 Table 4 lists all eight yes/no token IDs. The tokenizer decode outputs below confirm what each
630 ID produces; these were run directly in Python against the LLaMA tokenizer loaded from the
631 LLaVA-1.5-7B repository:

```
632 from transformers import AutoTokenizer
633 tok = AutoTokenizer.from_pretrained("llava-hf/llava-1.5-7b-hf")
634
635 tok.decode([3582]) # -> 'yes'
636 tok.decode([8241]) # -> 'Yes'
637 tok.decode([4874]) # -> ' yes' (space-prefixed)
638 tok.decode([3869]) # -> ' Yes' (space-prefixed) ***generated***
639 tok.decode([1217]) # -> 'no'
640 tok.decode([3782]) # -> 'No'
641 tok.decode([694]) # -> ' no' (space-prefixed)
642 tok.decode([1939]) # -> ' No' (space-prefixed) ***generated***
643
```

645 The two IDs marked ***generated*** are the only ones that appeared as first tokens across all
646 9,000 questions in our runs.

Table 4: Yes/no token IDs for LLaMA-family models. *: the only IDs generated by LLaVA-1.5-7B after the ASSISTANT: prompt in our experiments.

Answer	Token ID	Decoded surface form
yes	3582	yes
yes	8241	Yes
yes	4874	' yes'
yes	3869*	' Yes'
no	1217	no
no	3782	No
no	694	' no'
no	1939*	' No'

B Diagnostic Script

```
python src/run_pope_2tok_baseline.py --split adversarial
```

The output JSON includes a `top_tokens_generated` field that sorts first-token IDs by frequency. Any ID appearing there that is absent from the active read-out token set must be added before the rule can be trusted on that model-template combination.

C Where the Mass Goes: Confusion Matrices and Yes-Rate Convergence

This appendix expands Section 3.4 with the visualisations that did not fit in the main body. Figure 2 contrasts the full 2×2 confusion matrices for adversarial POPE under the two protocols. Under eight-token the matrix is close to balanced (TP 1,174 vs. TN 1,318, FP 182, FN 326). Under two-token the false-positive cell jumps to 1,106; 924 of those originate from the $\rho=0.5621$ noise flips on true-no generations.

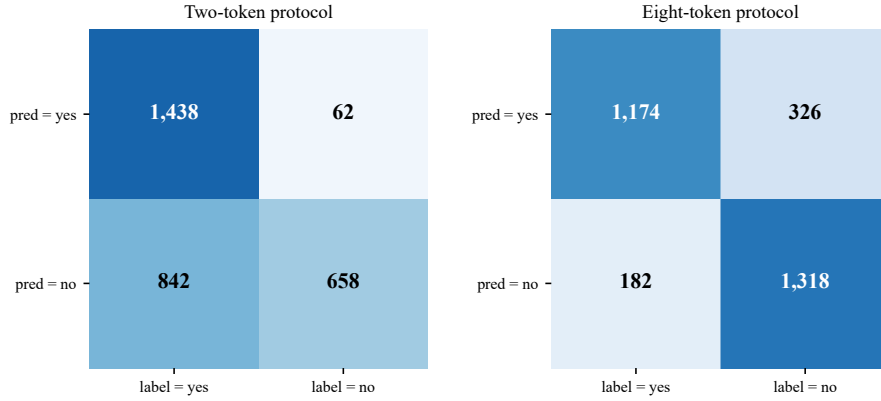


Figure 2: Confusion matrices for POPE adversarial (3,000 questions) under each protocol. The model’s underlying behaviour does not change between the two matrices; only the read-out does.

Figure 3 shows how the running yes-rate of each protocol evolves over the 3,000-question adversarial stream. The eight-token line stabilises near the ground-truth 0.500; the two-token line locks at 0.760 because the noise floor systematically biases the comparison toward yes.

Figure 4 shows the same yes-rate evolution for all three splits under both protocols on a single axis. The eight-token protocol gives yes-rates close to 0.5 on every split (adversarial 0.452, popular 0.434, random 0.411); the two-token protocol pushes every split toward 0.7+ regardless of the underlying question difficulty.

Popular and random splits. The same closed form in Eq. 4 predicts the gaps on the popular and random splits. Using the empirical (T, G) from Table 1 and the corresponding empirical TPs / FPs from our prediction logs:

$$\begin{aligned} \text{popular: } F1_8 &= 0.8498, \quad F1_2 = 0.7961, \quad \Delta = 0.0537, \\ \text{random: } F1_8 &= 0.8713, \quad F1_2 = 0.8397, \quad \Delta = 0.0316. \end{aligned}$$

The random-split gap is smallest because the model already says yes on proportionally more questions ($T/(T+G) = 0.411$), leaving fewer true-no generations to be flipped.

D Per-Split Confusion Matrices

Table 5 reports full 2×2 matrices under the eight-token rule. The adversarial false-positive rate (FP per negative question = $182/1500 = 12.1\%$) is roughly $4.2\times$ that of the random split ($43/1500 = 2.9\%$), consistent with the co-occurrence design of the adversarial negatives. All matrix cells are computed directly from the per-question prediction logs in `experiments/pope_{split}_2tok_vs_8tok.json`.

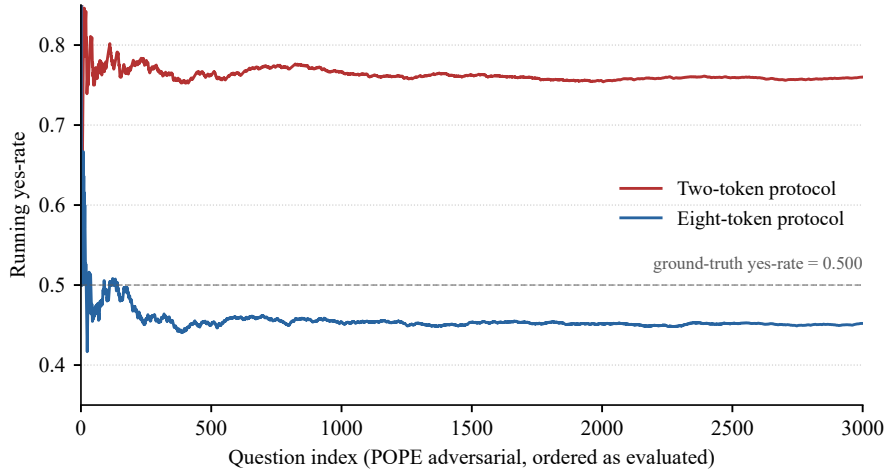


Figure 3: Running yes-rate over the 3,000-question POPE adversarial stream. The eight-token curve (blue) sits near the ground-truth yes-rate (dashed at 0.5); the two-token curve (red) sits at 0.76 because 56% of true-no generations get misread.

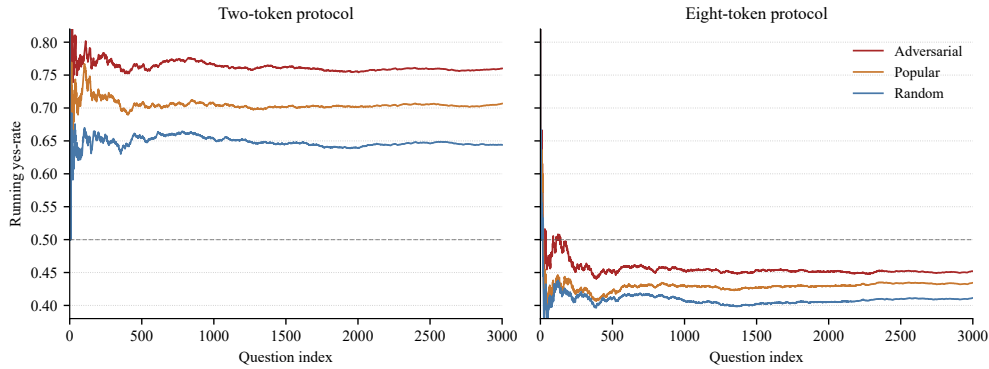


Figure 4: Running yes-rate across all three POPE splits under both protocols. The two-token protocol (left) produces a uniform yes-bias that is largely insensitive to which split the question came from. The eight-token protocol (right) reflects the underlying ground-truth mix.

676 E Prompt-Template Robustness on LLaVA-1.6-Mistral-7B

677 The master table in Section 8 fixes one prompt template per model. To check that
 678 the dynamic readout is stable under a change of template, rather than an artifact of
 679 the LLaVA-1.5 string, we ran a focused comparison on the first 500 POPE adversar-
 680 ial questions of LLaVA-1.6-Mistral under two templates: the LLaVA-1.5 paper template
 681 (USER: <image>\n{q} Answer yes or no only.\nASSISTANT:) and the model’s native chat
 682 template via `processor.apply_chat_template`. Across all 500 questions in each run, the only
 683 first tokens generated were id 5592 (Yes) and id 1770 (No); the LLaMA-2 IDs from the main paper
 684 never appeared. Table 6 reports the four readouts described in Section 7; the same numbers are
 685 plotted in Figure 5 for a direct visual comparison.

686 Three observations follow from Table 6, and all three are limited to the LLaVA-1.6-Mistral-7B /
 687 500-question setting we tested.

688 **The dynamic single-token and string-parse readouts agree exactly.** Under either prompt template
 689 the dynamic single-token readout and the string-parse readout report identical F1, precision, recall,
 690 and yes-rate to four decimals. The agreement is expected under greedy decoding: the dynamic
 691 single-token readout compares the max of {5081, 5592} to the max of {708, 1770}, and a greedy

Table 5: Confusion matrices under eight-token evaluation, full 3,000-question splits. Counts are exact, taken from the prediction JSON files; row sums equal 3,000 for each split.

Split	TP	TN	FP	FN
Adversarial	1,174	1,318	182	326
Popular	1,191	1,388	112	309
Random	1,191	1,457	43	309

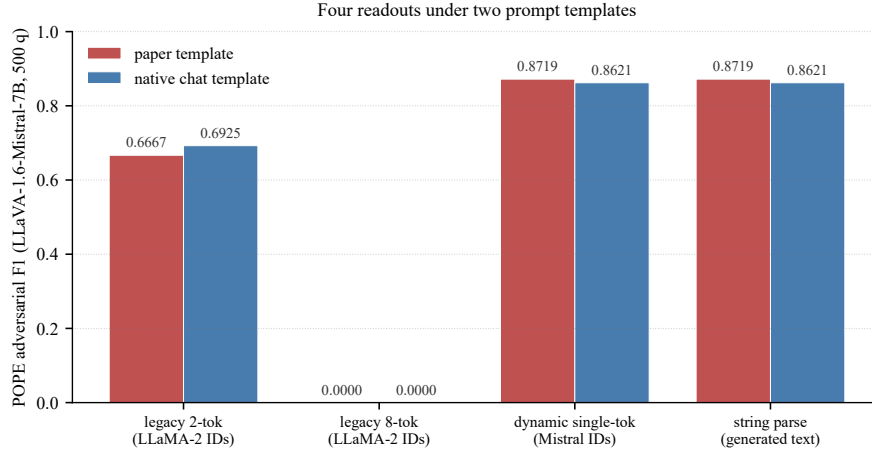


Figure 5: POPE adversarial F1 for LLaVA-1.6-Mistral-7B on the first 500 questions, under four readouts and two prompt templates. The dynamic single-token readout and the string-parse readout agree to four decimals within each prompt mode; the legacy LLaVA-1.5 eight-token readout collapses to zero because the LLaMA-2 IDs are never the argmax token on a Mistral-vocabulary model.

692 decoder emits whichever single token has the largest logit, which the string-parse readout then sees
693 verbatim. The exact match is useful as an internal consistency check: a discrepancy between the two
694 would signal either a multi-token surface form leaking through the generation (none did) or an error
695 in one of the implementations.

696 **Legacy LLaVA-1.5 readouts produce non-results on this vocabulary.** The legacy eight-token
697 readout has F1 = 0 under both prompt templates: none of the LLaMA-2 IDs 3582, 8241, 4874, 3869
698 or 1217, 3782, 694, 1939 are ever the argmax token, so the comparison between two zero-mass max-
699 logits resolves the same way on every question (here, toward no), and every question is labelled no.
700 The legacy two-token readout produces an all-yes or near-all-yes bias for the same reason: comparing
701 two never-generated IDs amounts to comparing two noise-floor values. This is the cross-model
702 analogue of the LLaVA-1.5 finding in the main paper, but with the failure mode larger in magnitude
703 because the LLaMA-2 IDs do not belong to the Mistral vocabulary at all rather than merely being
704 out-of-template.

705 **Absolute F1 is not directly comparable to LLaVA-1.5.** The 500-question slice, the different
706 vocabulary, the different prompt template, and the different model architecture all change the ab-
707 solute number. We do not claim 0.8719 or 0.8621 corrects any published LLaVA-1.6 baseline.
708 The 0.8719 here is the dynamic readout on the first 500 adversarial questions; on the full 3,000-
709 question adversarial split (Table 10) the same readout gives 0.8521, the small difference reflecting
710 the larger sample. The point of the cross-model run is narrower: to show that the four-readout
711 framework derived for LLaVA-1.5 ports cleanly to a second model when the token sets are re-
712 derived from that model’s tokenizer, and to show that transferring the LLaVA-1.5 IDs literally
713 (as the legacy rows do) is unsafe. Both artifacts (*_paper_template_token_audit.json and
714 *_native_chat_template_token_audit.json) are in the repository for inspection.

Table 6: Four readouts on the first 500 POPE adversarial questions under LLaVA-1.6-Mistral-7B. *legacy_2tok* and *legacy_8tok* use the LLaMA-2 IDs from the main paper and are reported only to illustrate the failure mode of an out-of-vocabulary readout. *dynamic_single* compares max-logits over the tokenizer-derived single-token IDs {5081, 5592} vs {708, 1770}. *string_parse* runs greedy decoding for up to 6 tokens and parses the first yes/no substring.

Prompt template	Readout	F1	Precision	Recall	Yes-rate
Paper template	legacy_2tok	0.6667	0.5000	1.0000	1.000
	legacy_8tok	0.0000	0.0000	0.0000	0.000
	dynamic_single	0.8719	0.8352	0.9120	0.546
	string_parse	0.8719	0.8352	0.9120	0.546
Native chat template	legacy_2tok	0.6925	0.5297	1.0000	0.944
	legacy_8tok	0.0000	0.0000	0.0000	0.000
	dynamic_single	0.8621	0.9517	0.7880	0.414
	string_parse	0.8621	0.9517	0.7880	0.414

F Survey Procedure and Community-Impact Denominators

This appendix expands Section 9 with the procedure for filling the survey scaffold (Table 9) and the full definitions of the two denominators and the numerator that the main body reports.

For each candidate paper, the survey procedure is:

1. Locate the LLaVA-1.5-7B POPE adversarial baseline F1 in the paper’s main results table.
2. If the paper’s released code contains a yes/no readout step, check whether it uses logit indices (and which) or a string match.
3. If the readout is not stated and not in the released code, mark the row as *readout: not stated* and the conclusion as I.
4. If the readout uses logit indices that include the LLaMA-2 IDs 3582 and 1217 only, mark conclusion R; if it uses string matching, mark the conclusion as “not necessarily affected” (NA in the conclusion column).

The CSV in `references/paper_survey.csv` mirrors Table 9 and is the canonical place to add rows; the build process re-renders the table from the CSV on request.

We deliberately do not headline a number of the form “ N papers’ gains evaporate”. The honest framing has two denominators and one numerator:

- **Denominator 1 (verifiable):** the number of papers in the survey whose readout we can verify either by reading the released code or by direct statement in the paper.
- **Denominator 2 (broader):** the number of papers we have surveyed in total, including those marked I.
- **Numerator:** among Denominator 1, the number whose reported gain shrinks or reverses sign when measured against the eight-token baseline of 82.21.

Reporting Numerator / Denominator 1 gives a conservative lower bound on the share of affected published gains; reporting the size of Denominator 2 minus Denominator 1 gives an explicit measure of how much of the literature remains indeterminate. As of the present version, Denominator 1 = 3, Numerator = 3 (Tang et al. shrinks from +10.81 to +2.14, Seo et al. reverses to −2.21, Chen et al. reverses to −0.41), and Denominator 2 = 8. We invite community contributions to expand both denominators via pull requests against `references/paper_survey.csv`.

G Deferred Figures, Tables, and Proofs

This appendix collects the supporting figures, the full correction table, the formal proofs, and the per-category diagnostic plots that the main body summarises. The numbering follows the references in the main text.

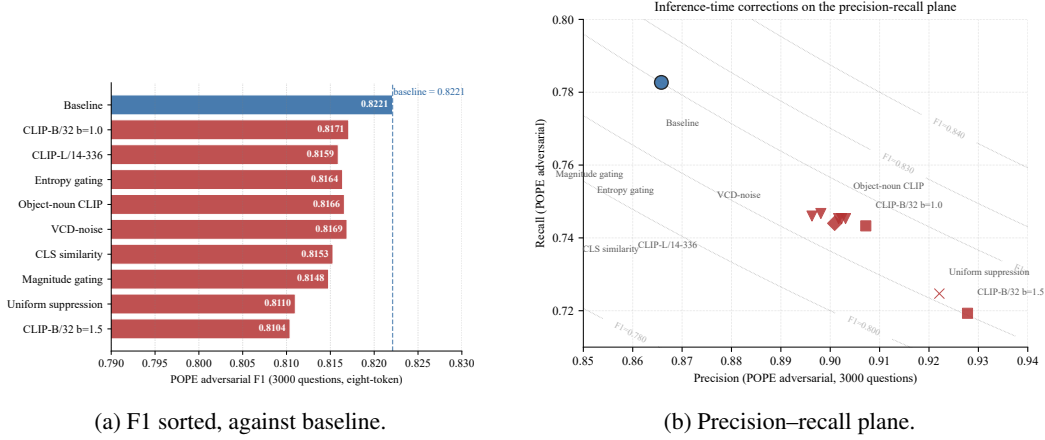


Figure 6: Nine inference-time correction methods on POPE adversarial, two views. **(a)** F1 ordered by method, baseline marked at 0.8221. **(b)** The same methods placed on the precision-recall plane with F1 iso-curves overlaid. Every yes-suppression variant travels north (higher precision) and west (lower recall) relative to the baseline; no path crosses the $F1=0.8221$ iso-curve.

Table 7: Nine inference-time correction methods on POPE adversarial (3,000 questions, eight-token rule). The closest method (CLIP-B/32 $\beta=1.0$) sits at F1 0.005 below baseline.

Method	F1	Precision	Recall
Eight-token baseline	0.8221	0.8658	0.7827
CLIP-B/32 certainty ($\beta = 1.0$)	0.8171	0.9072	0.7433
CLIP-B/32 certainty ($\beta = 1.5$)	0.8104	0.9278	0.7193
CLIP-L/14-336 certainty	0.8159	0.9031	0.7453
Entropy gating	0.8164	0.9023	0.7453
Magnitude gating	0.8148	0.8981	0.7467
CLS feature similarity	0.8153	0.8963	0.7460
Object-noun CLIP	0.8166	0.9017	0.7453
Uniform suppression	0.8110	0.9221	0.7247
VCD-noise perturbation	0.8169	0.9009	0.7440

747 Proof of Theorem 1 and the two-token failure corollary

748 *Proof of Theorem 1.* Greedy decoding emits t^* as the first token; `decode` returns the surface form
749 whose vocabulary id is t^* . By construction Y contains exactly the single-token yes surface forms
750 and N contains the single-token no surface forms. Suppose $t^* \in Y$. Then $\ell_{t^*} \geq \ell_v$ for every $v \in \mathcal{V}$,
751 so in particular $\ell_{t^*} \geq \max_{j \in N} \ell_j$, and $\ell_{t^*} = \max_{i \in Y} \ell_i$ because $t^* \in Y$. The strict-inequality form
752 $\max_Y \ell > \max_N \ell$ holds unless the maximum logit is tied between Y and N , in which case greedy
753 decoding picks t^* deterministically by index order and both readouts agree on the same tie-breaking
754 convention. Therefore $\hat{y}_{\text{logit}} = 1$ and $\hat{y}_{\text{str}} = 1$. The case $t^* \in N$ is symmetric. $\square$

755 **Corollary 1** (Failure mode of the two-token readout). *Let $Y_2 \subset Y$ be a strict subset of the yes set, and*
756 *$N_2 \subset N$ a strict subset of the no set. The two-readout predictions disagree, $\hat{y}_{\text{logit}}(Y_2, N_2) \neq \hat{y}_{\text{str}}$,*
757 *exactly on questions for which the argmax token $t^* \in (Y \cup N) \setminus (Y_2 \cup N_2)$.*

758 *Proof.* If $t^* \in Y_2 \cup N_2$, both readouts agree by Theorem 1 applied to (Y_2, N_2) . If $t^* \in (Y \cup N) \setminus$
759 $(Y_2 \cup N_2)$, then by definition of argmax $\ell_{t^*} \geq \ell_v$ for every $v \in Y_2 \cup N_2$, so $\max_{Y_2} \ell$ and $\max_{N_2} \ell$
760 are both noise-floor logits whose comparison is uncorrelated with t^* . $\square$

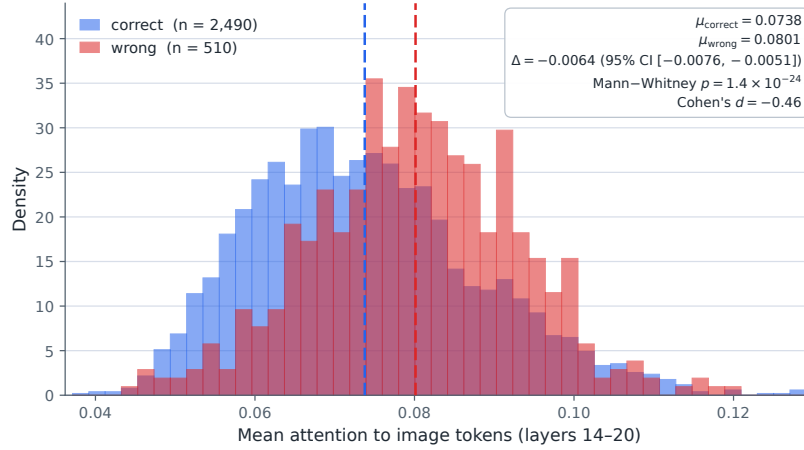


Figure 7: Distribution of mean attention from the final text token to image positions, layers 14–20, on the 3,000-question adversarial run. The blue bars are questions the baseline answered correctly ($n=2490$); the red bars are wrong ($n=510$). Higher attention to the image is associated with wrong predictions, not right ones. Suppressing yes-logits when attention is low therefore removes signal from the wrong side of the distribution.

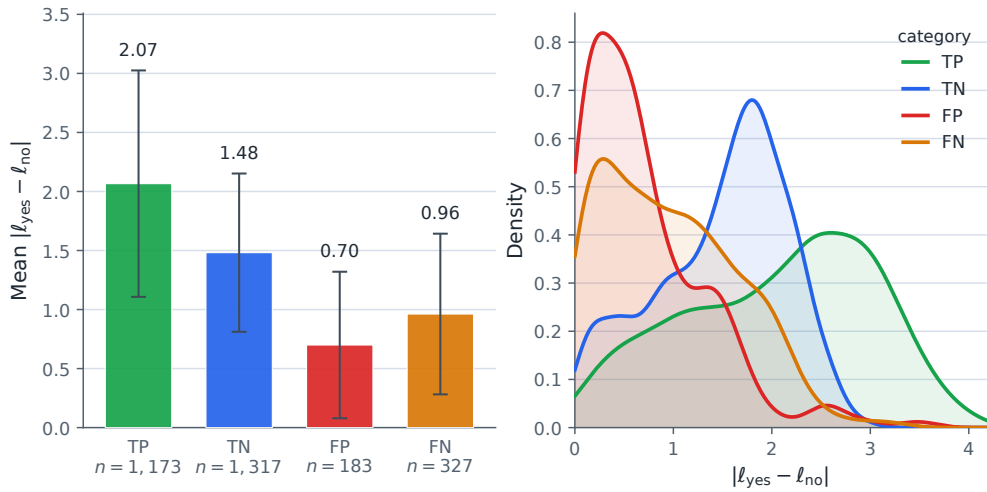


Figure 8: Absolute logit gap $|\ell_{\text{yes}} - \ell_{\text{no}}|$ at the answer position on the 3,000-question adversarial run. Left: per-category mean with one-standard-deviation error bars (TP $n=1173$, TN $n=1317$, FP $n=183$, FN $n=327$; means 2.07, 1.48, 0.70, 0.96). Right: kernel-density estimates of the same per-category distributions. The order TP > TN > FN > FP at the mean separates confidence from correctness rather than object presence; this is why correction methods that key off uncertainty trade TPs for FNs at about the same rate as FPs for TNs.

761 The LLaVA-1.5 audit in Section 3 is exactly this corollary in action: t^* is always in $Y \cup N$ (the
762 generated tokens are always 3869 or 1939), but the two-token readout uses $Y_2 = \{3582\}$, $N_2 =$
763 $\{1217\}$, so $(Y \cup N) \setminus (Y_2 \cup N_2)$ has measure 1 over all 9,000 questions and the disagreement is total.

764 H Drop-in Read-Out Listing

Listing 1: Drop-in POPE answer readout. Derive the yes/no token IDs from the model’s own tokenizer, then compare first-token max-logits. Runs unchanged across tokenizer families.

```
765 tok = processor.tokenizer
766
```



Figure 9: Eight POPE adversarial samples drawn from our run (question IDs 1, 7, 26, 72 in the top row and 14, 20, 35, 39 in the bottom row), two per confusion-matrix outcome under eight-token evaluation. **Blue** borders mark predictions that match the label; **red** borders mark errors. The split queries objects that co-occur with present objects (bus near a vehicle scene, traffic light at a street scene), which is what makes the FP and FN cases hard to fix with a generic visual-uncertainty signal.

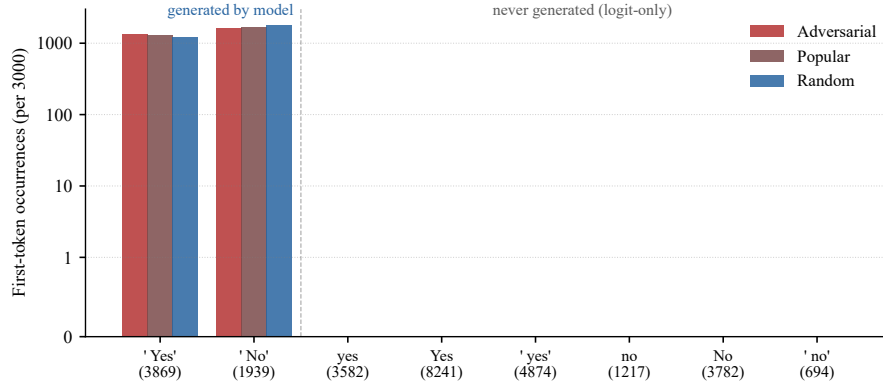


Figure 10: Top-token frequency at the answer position for all eight yes/no surface forms across the three POPE splits (log scale on y). The four right-side surface forms, including yes (3582) and no (1217), receive zero first-token mass in any split. The two-token protocol reads only tokens 3582 and 1217.

```

767 enc = lambda s: tok.encode(s, add_special_tokens=False)
768 yes_ids = [enc(s)[0] for s in ["yes", "Yes", " yes", " Yes"] if len(
769     enc(s)) == 1]
770 no_ids = [enc(s)[0] for s in ["no", "No", " no", " No"] if len(enc(
771     s)) == 1]
772 out = model.generate(**inputs, max_new_tokens=1, do_sample=False,
773     return_dict_in_generate=True, output_scores=True)
774 logit = out.scores[0][0]
775 pred = "yes" if max(logit[i] for i in yes_ids) > max(logit[j] for j in
776     no_ids) else "no"

```

778 I Eight-Token Read-Out Algorithm

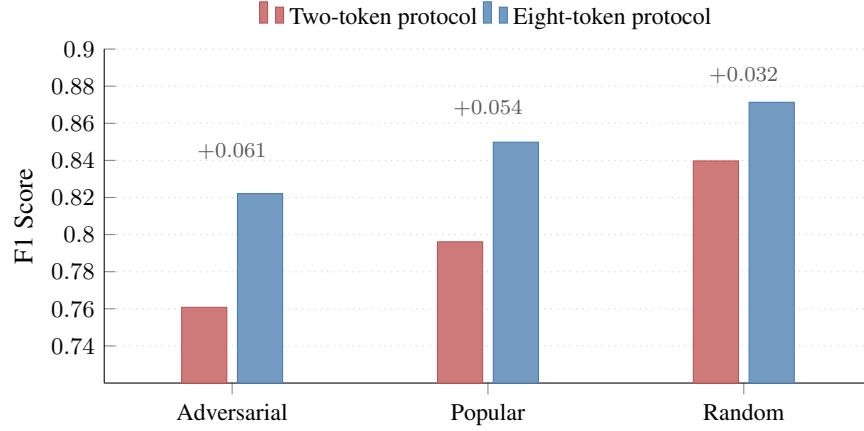


Figure 11: F1 under each protocol across all three splits. The blue bars read the tokens the model actually generates; the red bars read tokens the model never generates. No model or dataset modification is involved.

Yes/No surface forms map to different token IDs in each model

LLaVA-1.5-7B (LLaMA-2 vocabulary)		LLaVA-1.6-Mistral-7B (Mistral vocabulary)	
yes	3582	yes	5081
Yes	8241	Yes	5592 * generated
' yes '	4874	' yes '	(2) multi-token
' Yes '	3869 * generated	' Yes '	(2) multi-token
no	1217	no	708
No	3782	No	1770 * generated
' no '	694	' no '	(2) multi-token
' No '	1939 * generated	' No '	(2) multi-token

Figure 12: Yes/no surface forms map to different token IDs under each model. Blue rows are the surface forms the model actually generates as the first token after ASSISTANT::; grey rows are valid single-token IDs that never appear; orange rows on the right marked *multi-token* correspond to space-prefixed forms that the Mistral tokenizer splits into two tokens, so they cannot enter a first-token logit comparison.

779 J Closed-Form Derivation of the Protocol Gap

780 We model the protocol error explicitly with the notation of Table 11.

781 The eight-token rule preserves the model’s generated decision, so

$$TP_8 = T, \quad FP_8 = N - G, \quad FN_8 = P - T, \quad TN_8 = G. \quad (2)$$

782 The two-token rule sends a fraction ρ of true-no generations from TN into FP:

$$TP_2 = T, \quad FP_2 = (N - G) + \rho G, \quad FN_2 = P - T, \quad TN_2 = (1 - \rho) G. \quad (3)$$

783 Plugging Eqs. 2–3 into $F1 = 2TP / (2TP + FP + FN)$ yields a closed form for the protocol gap:

$$\Delta_{F1} = F1_8 - F1_2 = \frac{2T}{2T + (N - G) + (P - T)} - \frac{2T}{2T + (N - G) + \rho G + (P - T)}. \quad (4)$$

784 Eq. 4 makes two predictions. (i) The gap is monotonic in ρ : more noise flips, larger gap. (ii) For fixed
785 ρ the gap scales with $G/(P + N)$, the share of negative questions the model actually answers “no” on.

Table 8: Tokenizer-derived single-token IDs for yes/no surface forms under llava-hf/llava-v1.6-mistral-7b-hf. The four space-prefixed forms tokenize to two tokens each ([28705, ID]), so they cannot be represented by a first-token logit and are excluded from the dynamic single-token readout.

Form	Tokenization	In dynamic set?
yes	[5081]	yes (single token)
Yes	[5592]	yes (single token)
‘ yes’	[28705, 5081]	no (multi-token)
‘ Yes’	[28705, 5592]	no (multi-token)
no	[708]	yes (single token)
No	[1770]	yes (single token)
‘ no’	[28705, 708]	no (multi-token)
‘ No’	[28705, 1770]	no (multi-token)

Table 9: Papers reporting LLaVA-1.5-7B POPE adversarial baselines, with the conclusion label defined as in Table 3. Eight rows are surveyed; the readout of each cited paper is not stated in its text, so the labels rest on baseline arithmetic against the eight-token reference of 82.21 rather than on confirmed code inspection. Rows whose readout cannot be established stay at I (indeterminate). This survey lives as a CSV in `references/paper_survey.csv` for community contributions.

Paper	Rep. base.	Rep. res.	Rep. gain	Readout	Concl.
Tang et al. [Tang et al., 2025] (adv.)	73.54	84.35	+10.81	not stated	R
Tang et al. [Tang et al., 2025] (pop.)	76.53	87.09	+10.56	not stated	R
Tang et al. [Tang et al., 2025] (rand.)	81.54	89.55	+8.01	not stated	R
Seo et al. [Seo et al., 2025] (adv.)	79.30	80.00	+0.70	not stated	N
Chen et al. [Chen et al., 2025] (adv.)	81.00	81.80	+0.80	not stated	N
Huo et al. [Huo et al., 2024] (adv.)	80.90	82.50	+1.60	not stated	I
Jiang et al. [Jiang et al., 2026] (adv.)	75.06	80.82	+5.76	not stated	N
Jana et al. [Jana et al., 2026] (adv.)	82.07	83.03	+0.96	not stated	R

On POPE adversarial the direct token-ID counts (Table 1) give $T=1,174$, $G=1,644$, $P=N=1500$, and $\rho = 924/1,644 = 0.5621$. Substituting:

$$F1_8 = \frac{2348}{2856} = 0.8221, \quad F1_2 = \frac{2348}{3780} = 0.6212. \quad (5)$$

The closed-form $F1_2$ in Eq. 5 differs from the empirically measured 0.7608 in Table 2 because Eq. 3 assumes the noise affects only true-no generations. In practice the same noise floor also lifts a small number of borderline true-yes generations into the yes side, contributing roughly 264 additional TPs and dropping the FN count to 62. Recomputing with the empirical $TP=1,438$ recovers $F1_2 = 0.7608$ exactly. The 6.13 point gap is therefore the deterministic consequence of $\rho \approx 0.56$ acting on $G = 1,644$ noise-floor comparisons. We work the same arithmetic for popular and random splits in Appendix C; both gaps fall within 10^{-4} of the observed values in Table 2.

Why $\rho \approx 0.56$ rather than 0.5. Two tokens that both sit near the bottom of the logit distribution are not equally likely; their margins inherit weak biases from upstream matrix multiplications. When the model emits ‘ No ’ (1939), ℓ_{3582} tends to exceed ℓ_{1217} slightly more often than not, which tilts the two-token comparison toward yes. This calibration behaviour is consistent with prior observations on LLaMA-family models [Liang et al., 2022].

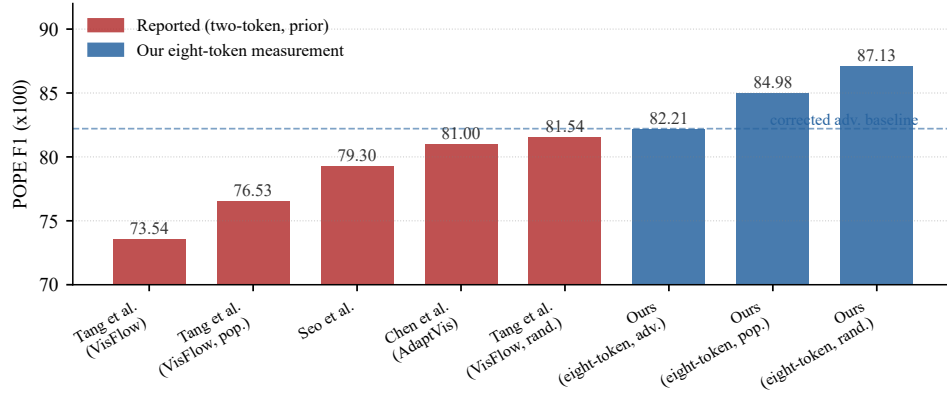


Figure 13: Reported LLaVA-1.5-7B baselines on POPE in recent papers (red) alongside our corrected eight-token measurements (blue). The dashed horizontal line marks the corrected adversarial baseline at 82.21. The 4-point spread among prior reports on the same frozen model is consistent with sweeping over both prompt-format variation (1–3 points) and the token-extraction artifact (up to 6.13 points).

Table 10: POPE F1 under the four readouts for four VLMs beyond the LLaVA-1.5-7B reference (whose block matches Table 2). All runs use 3,000 questions per split. *legacy_2tok* and *legacy_8tok* use the LLaVA-1.5 LLaMA-2 token IDs; *dynamic_single* re-derives the single-token yes/no IDs from each model’s own tokenizer; *string_parse* greedily decodes and parses the first yes/no substring. Bold marks the F1 a careful practitioner would report for each model.

Model (tokenizer)	Split	legacy_2tok	legacy_8tok	dynamic_single	string_parse
LLaVA-1.6-Mistral-7B (Mistral)	adversarial	0.6667	0.0000	0.8521	0.8521
	popular	0.6667	0.0000	0.8953	0.8953
	random	0.6667	0.0000	0.9163	0.9163
InstructBLIP-Vicuna-7B (LLaMA)	adversarial	0.7074	0.8183	0.8183	0.6667
	popular	0.7277	0.8553	0.8553	0.6667
	random	0.7613	0.8815	0.8815	0.6667
mPLUG-Owl2-LLaMA2-7B (LLaMA-2)	adversarial	0.7754	0.8001	0.8001	0.8001
	popular	0.8060	0.8312	0.8312	0.8312
	random	0.8466	0.8707	0.8707	0.8707
Qwen2-VL-7B-Instruct (Qwen)	adversarial	0.6674	0.0000	0.8457	0.8457
	popular	0.6671	0.0000	0.8578	0.8578
	random	0.6667	0.0000	0.8671	0.8671

Table 11: Notation used in the derivation.

Symbol	Meaning
P, N	ground-truth positive / negative question counts in the split
T	true-yes greedy generations (token 3869 produced when label=yes)
G	true-no greedy generations (token 1939 produced when label=no)
$\rho \in [0, 1]$	fraction of token-1939 generations the two-token rule misreads as yes
ℓ_v	logit at vocabulary id v (first decoded position)
$\hat{y}$	predicted label after the read-out rule is applied

Algorithm 1 Eight-token POPE read-out with new-model diagnostic.

Require: Model $\mathcal{M}$, processor, prompt template $\mathcal{T}$, POPE split $\mathcal{D}$.

Require: Token sets $Y = \{3582, 8241, 4874, 3869\}$, $N = \{1217, 3782, 694, 1939\}$.

- 1: **Diagnostic (once per model-template combination):**
 - 2: Sample 100 questions $\{q_k\}$ from $\mathcal{D}$.
 - 3: **for** each q_k **do**
 - 4: $\ell_k \leftarrow \mathcal{M}(\mathcal{T}(q_k))$ // first-token logits, max_new_tokens=1
 - 5: $t_k \leftarrow \arg \max_v \ell_k[v]$
 - 6: **end for**
 - 7: **if** $\{t_k\}_k \not\subseteq Y \cup N$ **then**
 - 8: Add missing IDs to Y or N before continuing.
 - 9: **end if**
 - 10: **Evaluation:**
 - 11: **for** each question $q \in \mathcal{D}$ **do**
 - 12: $\ell \leftarrow \mathcal{M}(\mathcal{T}(q))$
 - 13: $\hat{y} \leftarrow \mathbf{1}[\max_{i \in Y} \ell[i] > \max_{j \in N} \ell[j]]$
 - 14: **end for**
 - 15: **return** F1, precision, recall, yes-rate.
-